

Chapter 8

Image and Video Rendering and Assessment

IMP302m

Prerequisites

- 8.1 Image Quantization, Halftoning, and Printing
- 8.2 Perceptual Criteria for Image Quality Evaluation
- 8.3 Structural Approaches to Image Quality Assessment

8.1 Image Quantization, Halftoning, and Printing - Distortion Criteria

- Remind: effect of quantization in image compression
- Problem: Given an image $f(x, y)$ of size $M \times N$, where each pixel takes on values in the range $\{0, 1, \dots, D-1\}$, we want to quantize each image pixel to K output levels ($K < D$) so that a distortion criteria is minimized.
 - $K = 2$ if halftoning the image so that it can be printed using a binary laser or inkjet printer. In this case, the two output levels are 0 and $D-1$, respectively.
- The mean squared error criterion $D(x, y) = \sum_{x,y} (f(x, y) - b(x, y))^2$ where $b(x, y)$ is the quantized output

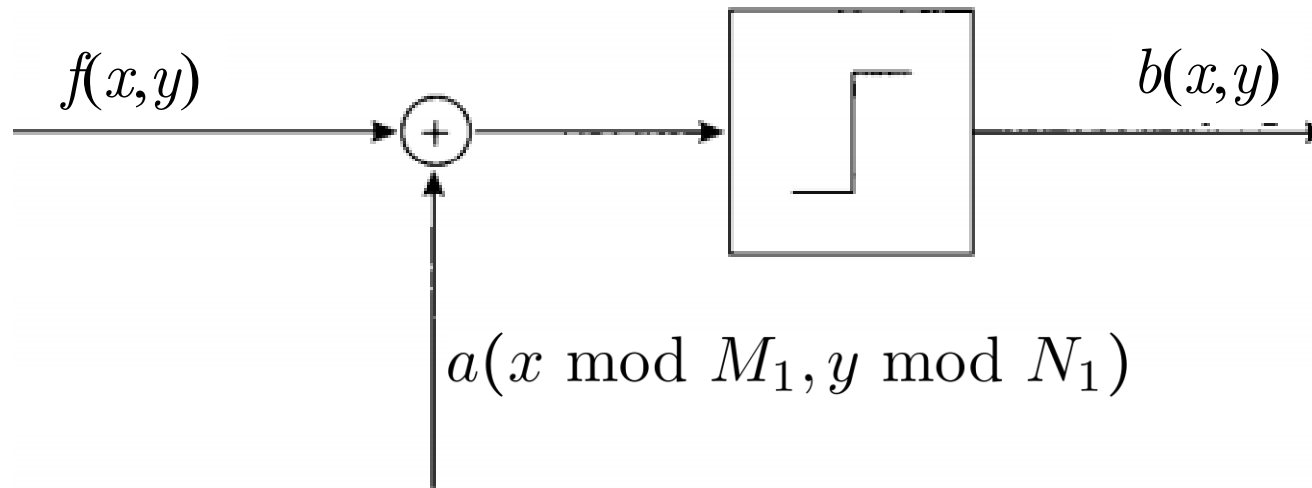
$$b(x, y) = Q(f(x, y)) = \begin{cases} D & \text{if } f(x, y) \geq D/2 \\ 0 & \text{otherwise} \end{cases}$$

8.1 Image Quantization, Halftoning, and Printing - Halftoning

- Halftoning is a process where a continuous tone image of, say 256 levels, is converted into an output image of 2 levels so that the original and the halftone images appear similar when observed from a distance.
 - The main application of halftoning and multitone is printing and in displaying images with limited bit-depth displays such as liquid crystal displays.
 - ordered dithering
 - error diffusion
 - optimization techniques
- lowest computational complexity
- 
- most computationally intensive

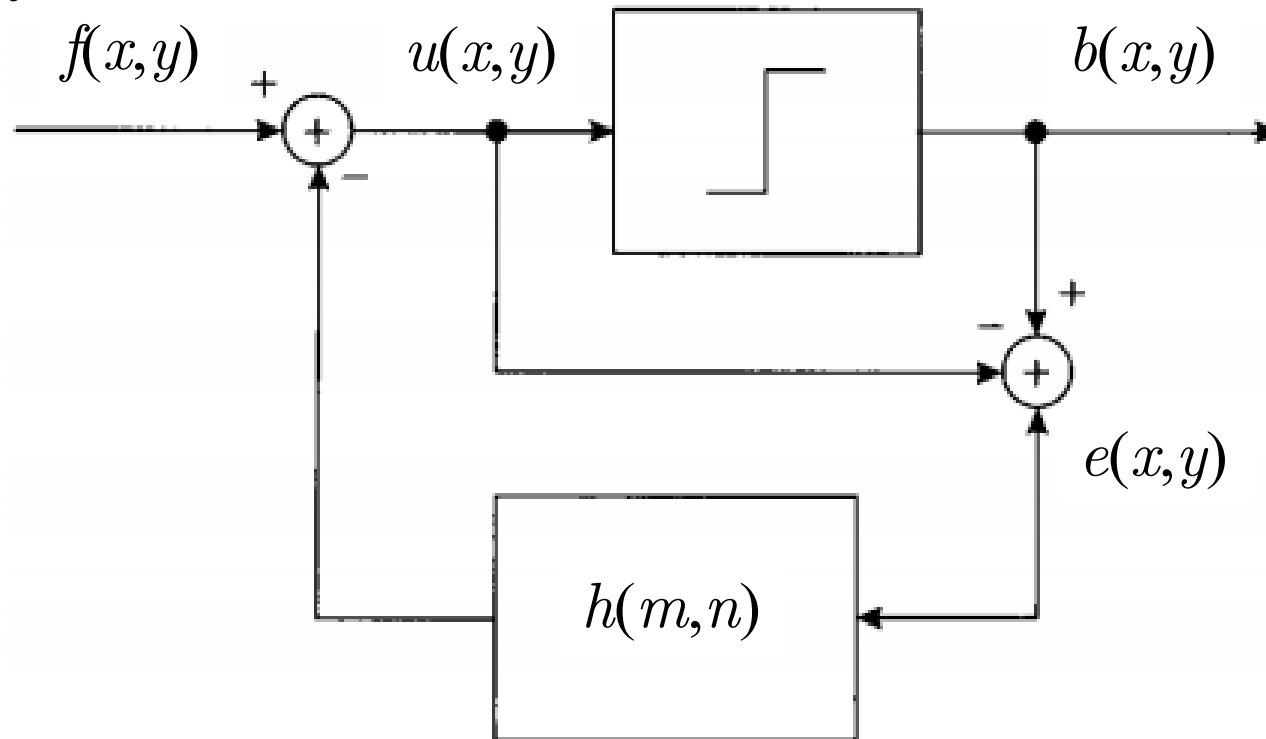
8.1 Image Quantization, Halftoning, and Printing - Halftoning

- Ordered dithering: An image $f(x, y)$ of size $M \times N$, where each pixel takes on values in the range $\{0, 1, \dots, D-1\}$, we want to generate a halftone $b(x, y)$ using ordered dithering



8.1 Image Quantization, Halftoning, and Printing - Halftoning

- Error Diffusion
- An excellent method for generating high quality halftones that are particularly suitable for low to medium resolution devices



8.1 Image Quantization, Halftoning, and Printing - Halftoning

- Error Diffusion

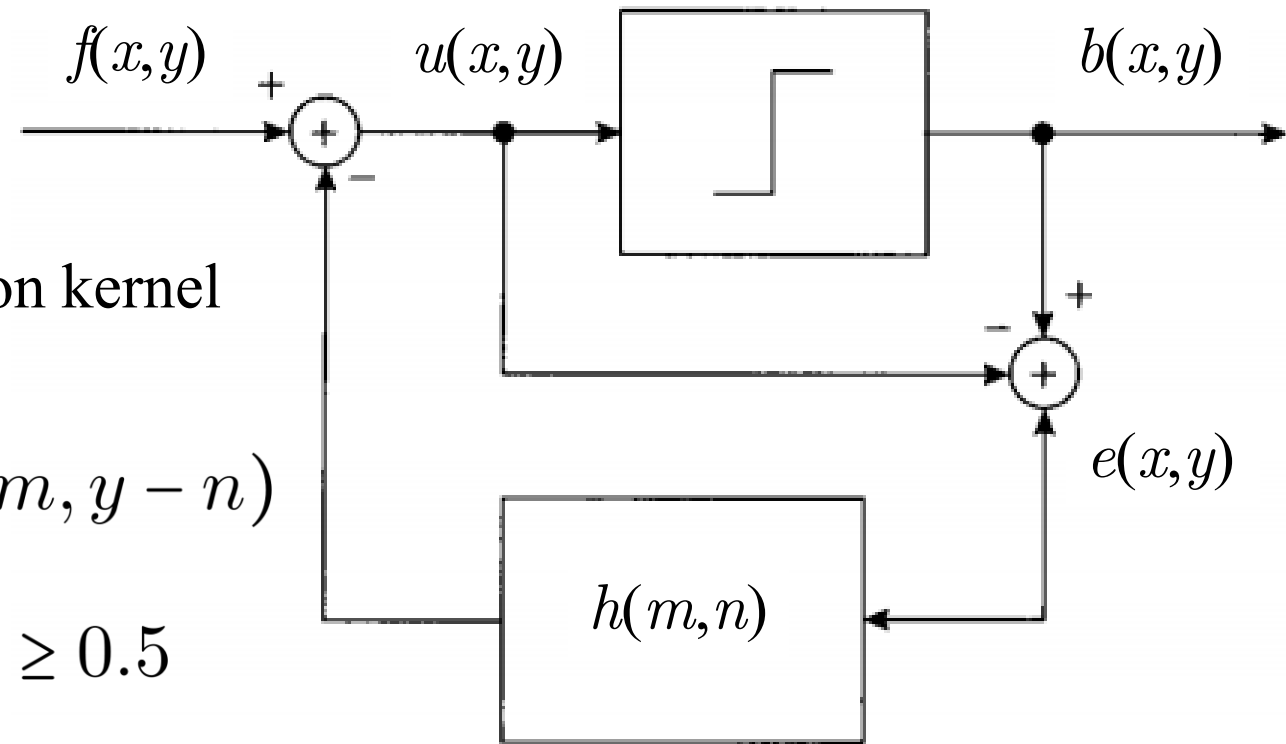
*state variable of the system
or modified input*

$$u(x, y) = f(x, y) - \sum_{m, n} h(m, n) e(x - m, y - n)$$

$$b(x, y) = Q(u(x, y)) = \begin{cases} 1 & \text{if } u(x, y) \geq 0.5 \\ 0 & \text{otherwise} \end{cases}$$

$$e(x, y) = b(x, y) - u(x, y) = Q(u(x, y)) - u(x, y)$$

$$\sum_{m, n} h(m, n) = 1$$



8.1 Image Quantization, Halftoning, and Printing - Halftoning

- Optimization techniques
- Consider a distortion criterion $d(f(x,y), b(x,y))$ between a continuous tone image $f(x,y)$ and a halftone $b(x,y)$
- The methods are

$$b^*(x,y) = \arg \min_{b(x,y)} \mathbb{E} \{d(f(x,y), b(x,y))\}$$

8.1 Image Quantization, Halftoning, and Printing - Color Quantization

- Problem: Consider the problem of displaying a color image $f(x,y)$ where each pixel is represented by K bits (e.g., $K = 24$ for RGB image). Suppose the display is only capable of handling L bits ($L < K$) or 2^L different colors $\rightarrow$ To display an image using such a device, we convert $f(x,y)$ to an image $c(x,y)$ where each pixel $c(x,y)$ is described by an L -bit index.
- The problem of color quantization is to design the color palette (hence the look-up table), and to determine how each pixel is to be mapped to the colors in the palette

8.1 Image Quantization, Halftoning, and Printing - Color Quantization

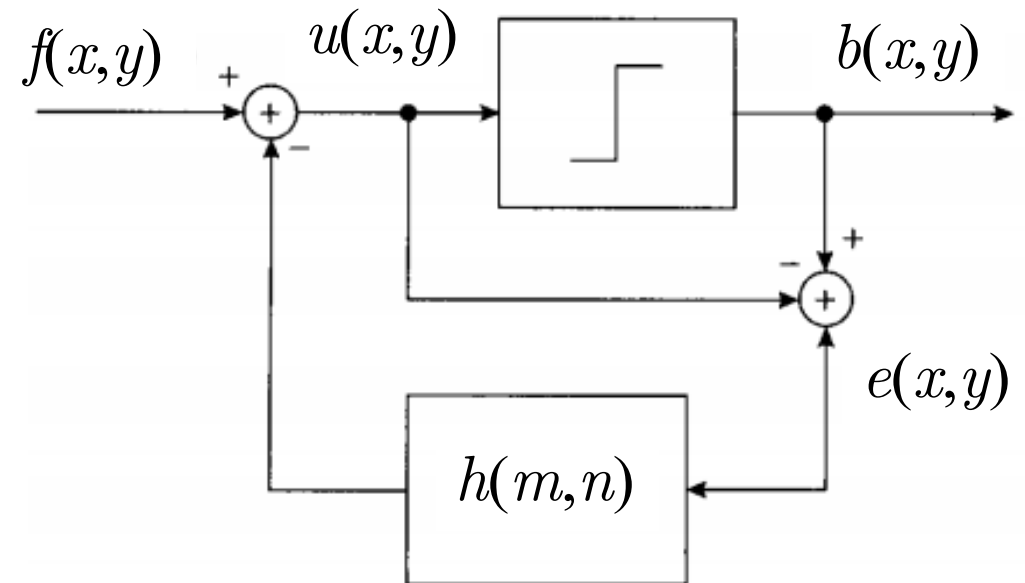
- **Color Palette Design**

- Median cut algorithm: each color within the color palette is used for representing approximately the same number of pixels in the original image $f(x,y)$
- Consider a 24-bit image $f(x,y) \rightarrow$ compute the color histogram of the image (a maximum of 16 million different colors) $\rightarrow$ “pre-quantize” the original image into a 15-bit (5-bit uniform quantizer for each color channel) $\rightarrow$ 32,768 colors after this step
- Find a rectangular box that tightly encloses the available colors within the image $\rightarrow$ The specific cut is made at the median point so that approximately equal number of pixels fall into each half of the sub-divided boxes

Repeat the step until reaching the desired number of colors

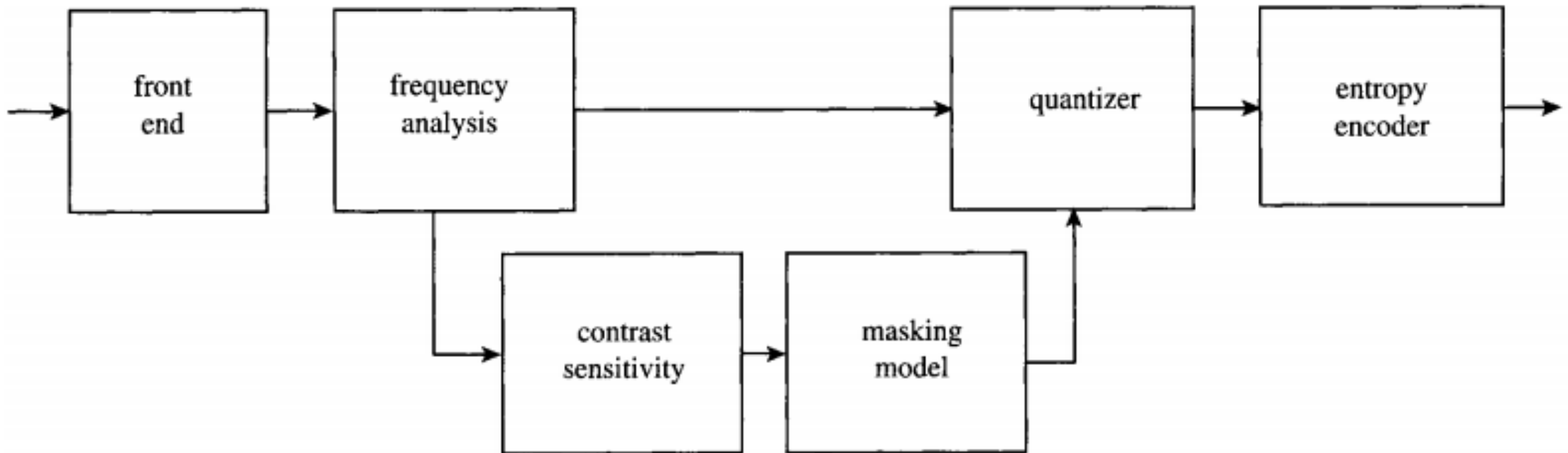
8.1 Image Quantization, Halftoning, and Printing - Color Quantization

- **Error Diffusion Approach** (after **Color Palette Design**)
- Nearest neighbor mapping: a straightforward approach, where we map each color pixel to the nearest color in the color palette
- Error diffusion: replace the one bit quantizer by a vector quantizer with the color palette as the code book



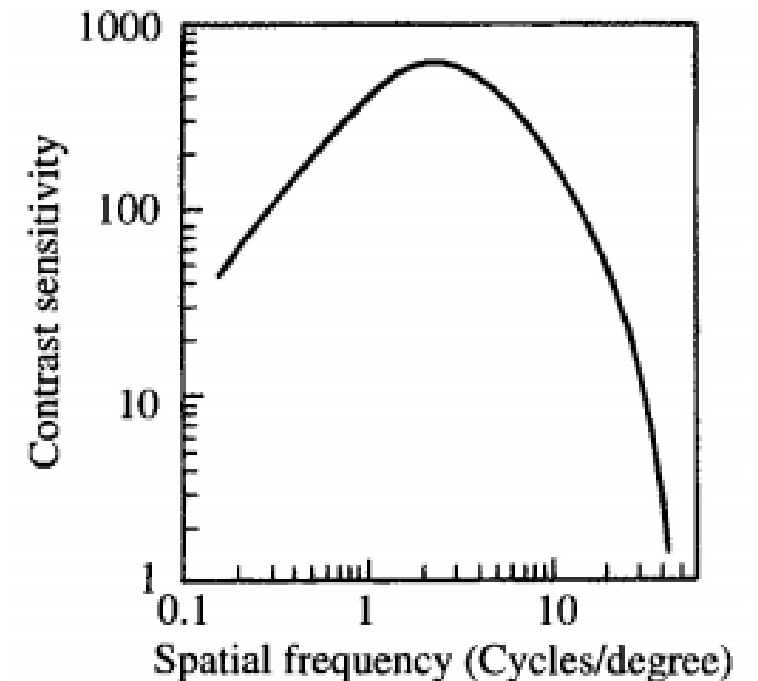
8.2 Perceptual Criteria for Image Quality Evaluation

- The perceptual models are based on properties of the visual system and measurements of the eye characteristics (the contrast sensitivity function (CSF), light adaptation, masking)



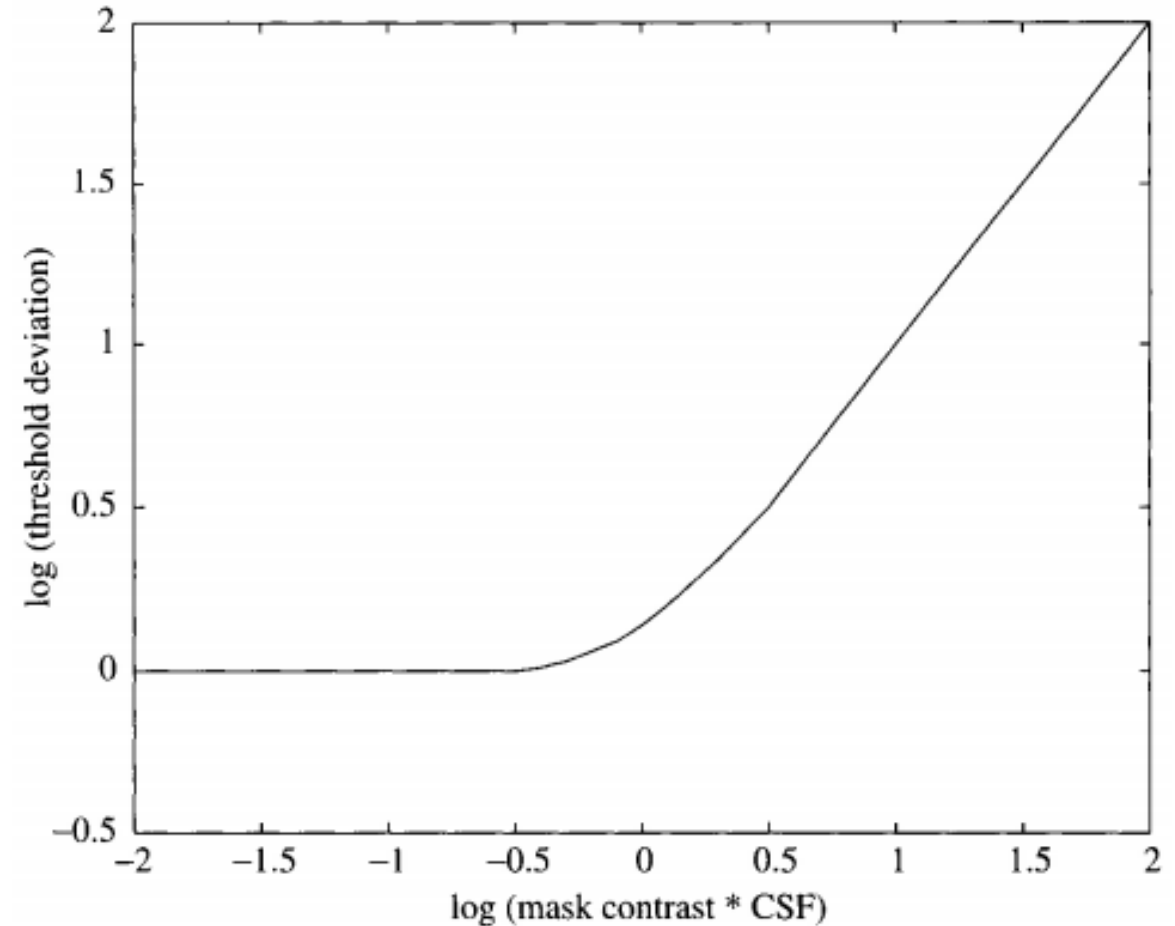
8.2 Perceptual Criteria for Image Quality Evaluation - Fundamentals of Human Perception

- **Psychophysics of Vision**
- Contrast Sensitivity Function: provides a characterization of its frequency response.
- The contrast sensitivity function can be thought of as a bandpass filter
- One of these methods involves the measurement of visibility thresholds of sine-wave gratings in a manner analogous to the experiment



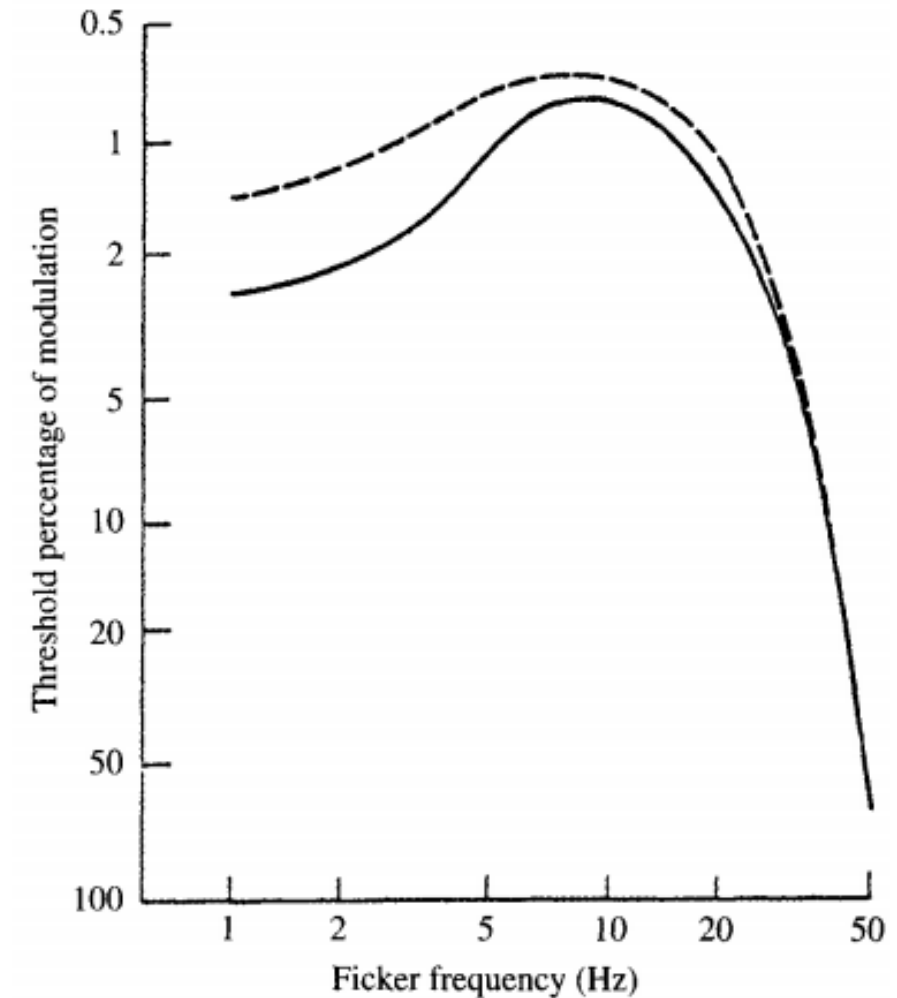
8.2 Perceptual Criteria for Image Quality Evaluation - Fundamentals of Human Perception

- **Psychophysics of Vision**
- Contrast masking refers to the reduction in visibility of one image component caused by the presence of another image component with similar spatial location and frequency content.



8.2 Perceptual Criteria for Image Quality Evaluation - Fundamentals of Human Perception

- **Psychophysics of Vision**
- Temporal masking: Temporal masking of time varying stimuli is extremely important in determining the quality of a video signal
- The perception of a moving object depends heavily on whether or not the object is being tracked by the eye
- The temporal contrast sensitivity function can be thought of as an extension to the spatial contrast sensitivity function



8.2 Perceptual Criteria for Image Quality Evaluation - Perceptual Image Quality Metrics

- The goal of a perceptual metric for image quality is to determine the differences between two images that are visible to the human visual system.
- Metrics
 - Calibration: The array of numbers that represents an image may have come from a number of different devices and could have gone through different transformations (conversion to densities, gamma correction, etc.) before it is displayed for observation by a human eye
 - Registration: the point-by-point correspondence between two images, is necessary for any quality metric to make any sense.
 - Display model: an essential part of any image quality metric as the human visual system can only see what the display can reproduce.

8.3 Structural Approaches to Image Quality Assessment

- The Minkowski metric between a reference image x (assumed to have perfect quality) and a distorted image y is defined as

$$E_p = \left(\sum_{i=1}^N |x_i - y_i|^p \right)^{1/p}$$

where x_i and y_i are the i -th samples in images x and y , respectively, N is the number of image samples, and p refers to the degree of power and varies in the range of $p \in [1, \infty)$

- Discussion: Is it really good?

8.3 Structural Approaches to Image Quality Assessment - Structural Similarity Index

- First, the luminance of each signal is estimated as the mean intensity

$$\mu_x = \bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$$

- The luminance comparison function $l(\mathbf{x}, \mathbf{y}) = l(\mu_x, \mu_y) = \frac{2\mu_x\mu_y + C_1}{\mu_x^2 + \mu_y^2 + C_1}$

- Second, use the standard deviation as an estimate of the signal contrast

$$\sigma_x = \left(\frac{1}{N-1} \sum_{i=1}^N (x_i - \mu_x)^2 \right)^{1/2}$$

- The contrast comparison $c(\mathbf{x}, \mathbf{y}) = c(\sigma_x, \sigma_y) = \frac{2\sigma_x\sigma_y + C_2}{\sigma_x^2 + \sigma_y^2 + C_2}$

$$C_1 = (K_1 L)^2, C_2 = (K_2 L)^2 \text{ where } K_1, K_2 \ll 1$$

8.3 Structural Approaches to Image Quality Assessment - Structural Similarity Index

- Third, the signal is normalized (divided) by its own standard deviation, so that the two signals being compared have unit standard deviation.

$$s(\mathbf{x}, \mathbf{y}) = s\left(\frac{\mathbf{x} - \mu_x}{\sigma_x}, \frac{\mathbf{y} - \mu_y}{\sigma_y}\right) = \frac{2\sigma_{xy} + C_3}{\sigma_x\sigma_y + C_3}$$

where $\sigma_{xy} = \frac{1}{N-1} \sum_{i=1}^N (x_i - \mu_x)(y_i - \mu_y)$ and $C_3 = (K_3L)^2$ where $K_3 \ll 1$

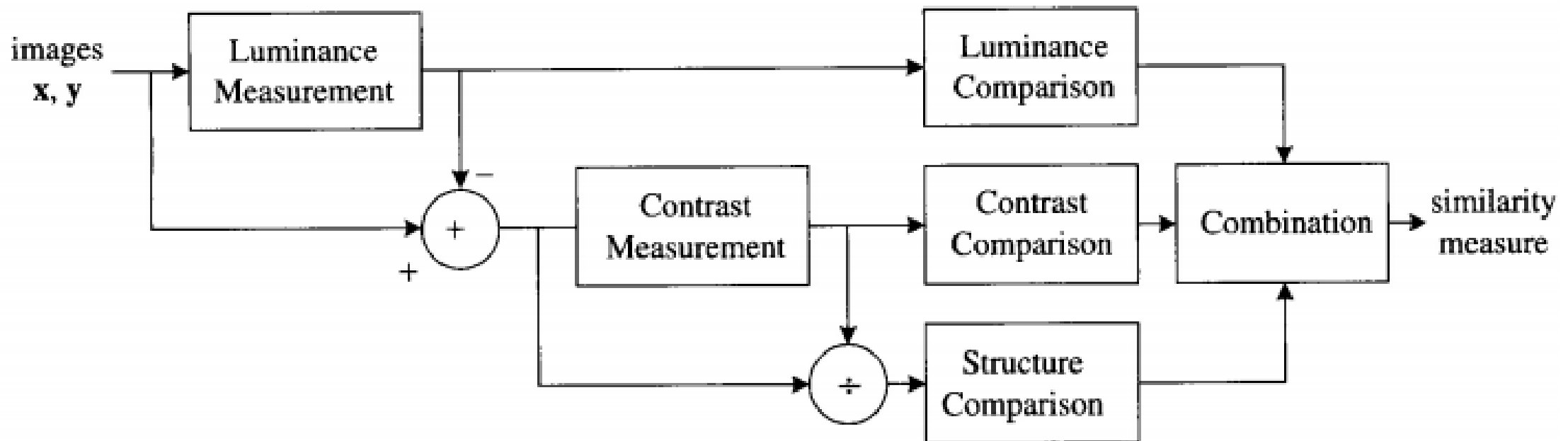
- Finally, the three components are combined to yield an overall similarity measure

$$S(\mathbf{x}, \mathbf{y}) = f(l(\mathbf{x}, \mathbf{y}), c(\mathbf{x}, \mathbf{y}), s(\mathbf{x}, \mathbf{y})) = [l(\mathbf{x}, \mathbf{y})]^\alpha \cdot [c(\mathbf{x}, \mathbf{y})]^\beta \cdot [s(\mathbf{x}, \mathbf{y})]^\gamma$$

- Important: three components are relatively independent

8.3 Structural Approaches to Image Quality Assessment - Structural Similarity Index

- Diagram of image similarity measurement system



8.3 Structural Approaches to Image Quality Assessment - Image-Quality Assessment

- A circular-symmetric Gaussian weighting function $\mathbf{w} = \{w_i | i = 1, 2, \dots, N\}$ with unit sum ($\sum_{i=1}^N w_i = 1$) is adopted.
- The estimates of local statistics are then modified accordingly

$$\mu_x = \frac{1}{N} \sum_{i=1}^N w_i x_i \quad \sigma_x = \left(\sum_{i=1}^N w_i (x_i - \mu_x)^2 \right)^{1/2} \quad \sigma_{xy} = \sum_{i=1}^N w_i (x_i - \mu_x)(y_i - \mu_y)$$

- The SSIM index between $\mathbf{X}$ and $\mathbf{Y}$ is defined as

$$\text{SSIM}(\mathbf{X}, \mathbf{Y}) = \frac{\sum_{j=1}^{N_s} W_j(\mathbf{x}_j, \mathbf{y}_j) \cdot \text{SSIM}(\mathbf{x}_j, \mathbf{y}_j)}{\sum_{j=1}^{N_s} W_j(\mathbf{x}_j, \mathbf{y}_j)}$$

where N_s is the number of samples in the quality map, and $W_j(\mathbf{x}_j, \mathbf{y}_j)$ is the weight given to the j -th sample.

Implementation

- Median cut algorithm for 8 levels, 16 levels, and 32 levels.
- Then evaluate the output image and the original one

Summary

- Image Quantization, Halftoning, and Printing
 - ✓ Distortion Criteria,
 - ✓ Halftoning,
 - ✓ Color Quantization
- Perceptual Criteria for Image Quality Evaluation
- Structural Approaches to Image Quality Assessment